

[← Go to NeurIPS 2026 Conference homepage \(/group?id=NeurIPS.cc/2026/Conference\)](/group?id=NeurIPS.cc/2026/Conference)

DUET: Principled Experience Replay for LLM Agent Reinforcement Learning

 (/pdf?id=jgomGjrmjA)

Qisheng Hu (/profile?id=~Qisheng_Hu1),
Quanyu Long (/profile?id=~Quanyu_Long1),
Shafiq Joty (/profile?id=~Shafiq_Joty1),
Wenya Wang (/profile?id=~Wenya_Wang1) 

 05 May 2026 (modified: 28 May 2026)  NeurIPS 2026 Conference Submission

 Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors

 Revisions (/revisions?id=jgomGjrmjA)

 CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>)

TL;DR: DUET makes teacher replay compatible with on-policy RL for LLM agents by correcting replay-induced biases and adaptively converting teacher trajectories into action and state-level guidance.

Abstract:

Experience replay from strong teacher agents is an appealing way to overcome the cold-start problem in on-policy RL for LLM agents. Yet in GRPO-style updates, inserting replayed teacher trajectories into the same rollout batch as student trajectories is not a benign form of supervision: successful teacher rollouts can contaminate the group baseline and suppress credit for rare successful on-policy student rollouts, while teacher actions remain off-policy when exact teacher--student importance correction is unavailable. Existing teacher-replay methods also tend to treat teacher trajectories mainly as action sequences to imitate or mix into the policy-gradient batch, leaving state-level progress signals largely unused. We propose DUET, a dual-channel framework for teacher experience replay. The Action Channel uses teacher actions for corrected action-level learning, combining asymmetric baseline separation with a discriminator-based density-ratio weight that approximates the missing teacher--student correction and attenuates teacher-action gradients as the student improves. The State Channel uses teacher observations to build a progress map over environment states and shapes rewards for on-policy rollouts, guiding exploration without requiring the student to copy teacher actions. Across ALFWorld and WebShop with two student model scales, DUET achieves the best performance in all used settings, improving the strongest prior baseline by 13.0 percentage points on average and by 17.5 points in the weakest-agent regime. Ablations show that each component contributes to DUET's performance, with the largest drop coming from removing baseline separation and complementary gains from density-ratio correction and state-channel shaping. Code is available at <https://anonymous.4open.science/r/DUET-DC26> (<https://anonymous.4open.science/r/DUET-DC26>).

Checklist Confirmation:  I confirm that I have included a paper checklist in the paper PDF.

Responsible Reviewing:  We acknowledge the responsible reviewing obligations as authors.

Primary Area: Deep learning advancements (e.g., architectures, optimizers, representation learning)

Secondary Area: Decision making, reinforcement learning, and control (e.g., hierarchical RL, multi-agent systems, agentic AI)

Contribution Type: General: Most submissions will fall into this type.

Academic Integrity:  I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (<https://neurips.cc/Conferences/2026/MainTrackHandbook> (<https://neurips.cc/Conferences/2026/MainTrackHandbook>)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.

LLM Usage:  Editing (e.g., grammar, spelling, word choice)

Declaration:  I confirm that the above information is accurate.

Reviewer Nomination:  Qisheng Hu (/profile?id=~Qisheng_Hu1)

Submission Number: 32282

Discussion (?id=jgomGjrmjA#discussion)

Filter by reply type... ▾

Filter by author... ▾

Search keywords...

Sort: Newest First

-

=

≡

👁

Everyone

Program Chairs

Submission32282...

Submission32282...

16 / 16 replies shown

Submission32282...

Submission32282...

Submission32282...

Submission32282...

Submission32282...

✕

Add:

Withdrawal

Official Comment

Author AC Confidential Comment

Official Review of Submission32282 by Reviewer y9x6

Official Review by Reviewer y9x6 📅 26 Jun 2026, 13:06 (modified: 29 Jul 2026, 03:07)
👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer y9x6
📄 Revisions (/revisions?id=Dpdf6TuYOp)

Summary:
This work addresses the challenge of using successful teacher trajectories in reinforcement learning for LLM agents in sparse-reward environments. In these settings, on-policy GRPO-style training struggles because the student rarely succeeds early in training, meaning there is little positive reward signal. The paper argues that mixing teacher trajectories into the same GRPO rollout group is problematic for two reasons: first, successful teacher rollouts can suppress the credit assigned to rare successful student rollouts; second, teacher actions are off-policy.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

The central framing is novel: teacher trajectories are decomposed into action-level supervision and state-level progress guidance. This is a more principled view than simply mixing teacher demonstrations into RL batches or using them as supervised fine-tuning data. The baseline-contamination argument is also a strong conceptual contribution. It identifies a concrete failure mode in teacher-mixed GRPO that is easy to miss but plausible in sparse-reward cold-start settings.

The method is technically coherent. Baseline separation directly addresses the group-normalisation issue. Density-ratio repair is a reasonable approximation when exact teacher likelihoods are unavailable. The State Channel is also intuitively well-motivated: successful trajectories contain intermediate progress states, not just final actions.

Weaknesses

The largest concern is generalisation. DUET is evaluated on ALFWorld and WebShop, which are both structured text-based environments. The State Channel depends on matching student states to teacher-visited states. This may work well when states are textual/discrete but it is less clear whether the method would transfer to noisier, more open-ended, partially observable environments.

A second concern is the density-ratio repair mechanism. The discriminator-based estimate is only an approximation of the missing teacher-student correction. Since the teacher cache is success-filtered, the learned discriminator may capture not only policy mismatch, but also trajectory-quality differences between successful teacher rollouts and student rollouts. As a result, it is unclear whether DR3 should be interpreted as a principled density-ratio estimator or as a learned heuristic weighting mechanism. This does not undermine the empirical usefulness of DR3, but it weakens the strength of the paper’s theoretical interpretation.

Minor concerns The behaviour-cloning component is adaptive, but it still provides a strong source of direct imitation supervision. Therefore, an important question is whether the observed gains arise primarily from the proposed replay-correction framework or from the additional imitation signal itself. In other words, does DUET

outperform prior methods because of its separation and correction mechanisms, or because it receives more effective supervised guidance from teacher trajectories? Additional analysis isolating these effects would strengthen the paper's central claims.

The method depends on a successful teacher trajectory cache. The paper could more clearly explain how sensitive DUET is to teacher quality, cache diversity, cache size, and the success-filtering process.

The comparison to broader off-policy RL or replay methods could be expanded. The current baselines are relevant, but additional comparisons or discussion would help position DUET more strongly.

Quality: 3: good

Clarity: 3: good

Significance: 3: good

Originality: 2: not good

Questions:

1. The paper appears to report results from a single training run due to computational cost. Given the high variance often observed in sparse-reward RL for LLM agents, how robust are the reported improvements across random seeds? Could the authors provide multi-seed results on at least a representative subset of settings, or report confidence intervals, to demonstrate that the gains are statistically reliable rather than artifacts of training variance?
2. How sensitive is DUET to the number of teacher trajectories and the teacher-student mixing ratio?
3. How large is the teacher trajectory cache, and how diverse are the successful trajectories within it?
4. A cache-size ablation would be useful: how does DUET perform with small, medium, and large teacher caches?
5. A teacher-quality ablation would be useful: what happens if the teacher is only moderately stronger, noisy, or partially suboptimal?
6. A comparison against simpler reward-shaping baselines would help isolate the value of using teacher-derived progress maps.
7. A comparison against alternative off-policy correction methods would help evaluate whether DR3 is the best choice.

Limitations:

DUET assumes access to a successful teacher trajectory cache. This is a meaningful assumption, because collecting and filtering high-quality teacher rollouts may be expensive. It also assumes that state matching is reliable. This is plausible in ALFWorld and WebShop, but may be harder in visual, real-world embodied settings.

DUET introduces additional system complexity. The discriminator must be trained and synchronised, density-ratio weights must be computed and stabilised, and the State Channel requires a progress map. In larger-scale distributed RL systems, these extra moving parts may introduce engineering overhead and new failure modes.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

No

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Add:

Official Comment

Author AC Confidential Comment



Official Comment by Authors

Edit ▼



Official Comment

by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))

03 Aug 2026, 20:16

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer y9x6

Comment:

Dear Reviewer y9x6,

As the discussion period is coming to a close, we would be grateful if you could review our response and the additional evidence provided. We have addressed each concern raised in your review; if any questions remain, we would sincerely appreciate your feedback and the opportunity to clarify them before the discussion ends. Thank you again for your time and thoughtful review.

Add:

Official Comment

Author AC Confidential Comment

Response for Reviewer y9x6, Part 1 of 2

Edit

Official Comment

by Authors (👤 Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))

📅 30 Jul 2026, 20:33

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer y9x6

Comment:

We sincerely thank the reviewer for the thorough and constructive review. We spent much of the rebuttal period on the experiments it called for, and the generalisation concern listed first in fact produced our best new result: **under 30% observation noise, DUET reaches 54.5%, above our own noise-free setting (47.5%)**. Given the number of points, we reply in two comments: Weaknesses 1–3 here, and Weaknesses 4–5 together with the questions in Part 2.

Weakness 1: transfer to noisier, more open-ended, partially observable environments.

We agree that the original exact-matching implementation may be brittle when observations contain surface-form variation or noise. To test whether this is a fundamental limitation of the State Channel or simply a limitation of the matching function, we introduced controlled corruption into the observations used by the matcher.

- **Noise:** each word of the state matcher's copy of the observation is independently dropped with probability 0.3 (word dropout); the policy network's input is left untouched. Task difficulty is therefore unchanged, and the only thing perturbed is the state-matching step;
- **Soft matching:** the default exact-hash lookup is replaced by TF-IDF text-similarity retrieval over the same progress table, taking the most similar stored state above a threshold.

val@100	exact matching	soft matching
clean observations	47.5 (paper setting)	51.5
30% observation noise	11.0	54.5
SC removed	31.0	

Exact matching is brittle under corruption, while soft matching preserves performance and does not sacrifice performance on clean observations. This suggests that the sensitivity lies in the matching operator, not in the use of teacher-state progress itself. Exact matching is therefore an implementation choice rather than a requirement of DUET.

Weakness 2: a principled density-ratio estimator, or a learned heuristic?

At the theory level, discriminator-based density-ratio estimation is well established. Under a Bayes-optimal discriminator, $D/(1 - D)$ recovers the exact likelihood ratio. For off-policy correction, the relevant denominator is the *actually replayed* distribution, namely the success-filtered cache, which is exactly what the discriminator is trained against. We also use a smoothed, bounded form, so teacher samples can only be down-weighted.

The deeper concern is whether success filtering makes the discriminator capture trajectory quality rather than policy mismatch. We designed a supplementary probe experiment to test this directly.

The setup: using the same features consumed by the discriminator, we train probes to separate teacher from student trajectories under two conditions:

- all student rollouts;
- successful student rollouts only.

If the features mainly encode success, accuracy should collapse in the second condition because both sides then contain only successful trajectories. Instead, accuracy **rises** from 84.7% to 90.0% on ALFWorld, and from 94.0% to 99.0% on late-stage WebShop, where the student already succeeds on 84% of rollouts. As a further check, successful and failed student trajectories receive nearly identical student probabilities (0.861 vs. 0.869), while teacher trajectories receive 0.280.

These results show that the features track *which policy produced the trajectory*, not *whether it succeeded*. DR3 therefore reflects policy mismatch rather than trajectory quality.

Weakness 3: Do the gains come from the correction mechanisms or the imitation signal?

We conduct two complementary analyses that point to the correction mechanisms. The main comparison uses ALFWorld 1.5B with the same training budget and curriculum:

Imitation setting	val@50	val@100
DUET (adaptive μ)	42.5	47.5
Imitation removed ($\mu = 0$)	40.5	34.0
GRPO (no teacher)	16.5	1.0

- 1. Removing imitation still retains most of the gain.** Without the imitation term, DUET reaches 34.0 versus GRPO's 1.0 on ALFWorld, and 16.5 versus 0.5 on WebShop. Thus, the correction mechanisms are effective on their own, while the remaining gap to full DUET reflects the additional benefit of adaptive imitation.
- 2. The training dynamics support the same conclusion.** The adaptive weight μ reaches its floor by step 30, when success is still near zero. Most of the subsequent improvement therefore occurs while imitation is already at its lowest weight. If imitation were the main driver, gains should emerge when its weight is highest, which is not what we observe.

Add:

Official Comment

Author AC Confidential Comment



Response for Reviewer y9x6, Part 2 of 2

Edit ▾

Official Comment
by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))
 30 Jul 2026, 20:37
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer y9x6

Comment:
Weakness 4: Sensitivity to teacher quality, cache diversity, and success filtering

We agree that these sensitivities deserve a clearer account. Most importantly, we ran a new teacher-quality ablation on ALFWorld 1.5B, replacing the 72B teacher with a substantially weaker 14B teacher while keeping the rest of the training setup unchanged.

Dimension	Setting	Result
Teacher quality	72B teacher (80.6% success)	47.5%
	14B teacher (68.1% success)	34.5%
Cache diversity	distinct seqs 8.5 demos/task	4.4; 7.6% single-path tasks

Dimension	Setting	Result
Success filtering	raw samples retained	80.6%
The new 14B experiment shows graceful degradation. Reducing teacher success by 12.5 points lowers DUET by 13 points, but DUET with the weaker teacher still reaches 34.5%, outperforming every baseline using the strong 72B teacher, including CHORD at 27.0%. The 14B run is also still improving at the end of the fixed budget, suggesting that weaker supervision slows learning rather than causing the mechanism to fail. The remaining statistics clarify the cache assumptions. The cache averages 4.4 distinct action sequences per task, with only 7.6% of tasks having a single path, while the success filter retains 80.6% of raw rollouts. Together, these findings suggest that DUET does not rely on an unusually diverse cache or an aggressively filtered set of demonstrations.		

Weakness 5: Comparison to broader off-policy RL and replay methods

We appreciate this suggestion and will expand the discussion of AWAC/IQL-style advantage weighting, V-trace truncation, and prioritized replay. These methods typically require quantities unavailable in our setting, such as teacher likelihoods or tokenizer-aligned action probabilities. Two controlled comparisons already isolate the value of our design: removing DR3 slows performance by 14 points at step 50, while LUFFY's $\pi/(\pi + \beta)$ weighting reaches only 5.5% at 1.5B and becomes net-negative at 7B.

Multi-seed robustness and confidence intervals

We ran six independent seeds on the representative ALFWorld 1.5B setting, using the same pinned curriculum:

val@100	Result
DUET ($n = 6$)	44.3 ± 7.1 (range: 34.5–53.0)
Strongest baseline, SFT→GRPO	30.0
CHORD / LUFFY / GRPO	27.0 / 5.5 / 1.0

Two observations are important. First, **the worst DUET seed, at 34.5%, still exceeds the strongest baseline at 30.0%.** Second, the learning dynamics are consistent across seeds: **five of six seeds improve during the second half of training by 4.5 to 13.5 points, while the remaining seed declines by only 3.5 points.** By comparison, all three baselines decline over the same interval (CHORD: 3.0 points; GRPO: 15.5; LUFFY: 20.5). Thus, DUET's advantage is not tied to a favorable seed or a single evaluation checkpoint.

How large is the teacher trajectory cache, and how diverse are the successful trajectories within it?

On ALFWorld, the cache contains 19,497 successful trajectories over 2,348 tasks, averaging 8.3 demonstrations per task. Among the 8.5 sampled demonstrations per task, 4.4 have distinct action sequences on average, and only 7.6% of tasks contain a single solution path. Thus, the reported gains do not rely on a highly diverse cache with many distinct solutions for every task.

What happens if the teacher is only moderately stronger, noisy, or partially suboptimal?

We completed the moderately weaker-teacher case by replacing the 72B teacher with a 14B teacher while keeping the remaining setup fixed. Teacher sample success decreases from 80.6% to 68.1%, while DUET decreases from 47.5% to 34.5%. The weaker-teacher result still exceeds CHORD using the strong 72B teacher at 27.0%, and the run is still improving at the end of the fixed budget. This shows graceful degradation under a moderately weaker teacher. We do not claim results for noisy or deliberately suboptimal teachers.

Could the authors compare against a simpler reward-shaping baseline?

We added a control that preserves the strength of the shaping signal while removing the teacher-derived progress structure. Within each task, we randomly permute the progress values among the same states, keeping both bonus magnitude and state coverage unchanged.

State Channel variant	val@100
True teacher-derived progress	47.5
Randomly permuted progress	41.0
State Channel removed	31.0

Generic shaping of the same strength improves over removing the State Channel, but remains 6.5 points below the true progress map. This isolates the contribution of the teacher-derived ordering, rather than merely the presence of an additional dense reward.

Add: [Official Comment](#) [Author AC Confidential Comment](#)

Official Review of Submission32282 by Reviewer bDeY

Official Review by Reviewer bDeY  25 Jun 2026, 06:46 (modified: 29 Jul 2026, 03:07)
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer bDeY
 Revisions (/revisions?id=9p7KTfLurb)

Summary:

This paper introduces a combination of multiple steps to use teacher data in LLM training. Specifically, it introduces Baseline Separation, DR3, Adaptive Behavior Cloning, and the State Channel. The combination leads to good performance on 2 benchmarks where state matching (required by the State Channel) is readily feasible.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

- The paper is clear and easy to follow.
- The idea of training a discriminator to estimate the importance ratio between student and teacher rollouts is interesting to me.
- The experimentation results show meaningful improvements in the selected domains.
- The ablations show that all components are useful in the proposed method.

Weaknesses

Claim about being "principled"

- Several important components appear heuristic rather than theoretically derived. In particular, the State Channel uses environment-specific state matching and trajectory-level reward shaping that the authors acknowledge is not policy-invariant. It is also crucial empirically: without it, performance is comparable to SFT+GRPO on ALFWorld (31.0% versus 30.0%) but substantially worse on WebShop (1.0% versus 18.5%). Thus, it is unclear whether the results support the proposed principled corrections or primarily reflect the benefit of a strong handcrafted shaping signal.
- DR3 might also be a bit questionable on whether it is "principled". The discriminator weight $\hat{w} = D/(1 - D) \approx \pi_\theta/\pi_\beta$ is already the full off-policy correction for a teacher sample. Why does Eq. 9 then also multiply by $\rho_t = \pi_\theta/\pi_{\theta_{old}}$? That makes the weight $\pi_\theta^2/(\pi_\beta\pi_{\theta_{old}})$, double-counting π_θ .

Breadth of the evaluation and potential limitation of the applicability

ALFWorld and WebShop are appropriate benchmarks, but the empirical scope is narrow and unusually favorable to the proposed State Channel, since both provide structured textual states amenable to handcrafted matching. It may require substantial adaptation to work in domains where exact state matching is hard, such as math or other reasoning tasks. The methods it compares to, such as CHORD, are more generally applicable.

Quality: 2: not good
Clarity: 3: good
Significance: 2: not good
Originality: 2: not good

Questions:

- What are the actual group composition numbers (n_o on-policy and n_β teacher per prompt)? Is n_β a fixed number, i.e. keep generating teacher rollouts until n_β successful rollouts are collected?
- In the SFT-GRPO baseline, does SFT use the exact same amount of data from the teacher? I assume yes if my understanding of "cache" is correct, because the paper says "the same teacher cache is used for all methods that require teacher trajectories." Can you maybe add the SFT training curve showing performance w.r.t. optimization epoch/step so that it is easy to evaluate whether SFT is properly executed.
- Can you add more details about the rollouts collected from the teacher? Like the size of the teacher cache, the average number of teacher rollouts per prompt, etc.

Limitations:

The authors' own limitation section has discussed a decent number of limitations that I can think of.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

1. Not all "the strongest non-DUET baseline" is underlined in Table 1. The underlines are missing for the 3B experiments.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Add: **Official Comment** **Author AC Confidential Comment**



Official Comment by Authors

Edit

Official Comment
by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))
 03 Aug 2026, 20:16
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer bDeY

Comment:
Dear Reviewer bdey,

As the discussion period is coming to a close, we would be grateful if you could review our response and the additional evidence provided. We have addressed each concern raised in your review; if any questions remain, we would sincerely appreciate your feedback and the opportunity to clarify them before the discussion ends. Thank you again for your time and thoughtful review.

Add: **Official Comment** **Author AC Confidential Comment**



Official Comment by Authors

Edit

Official Comment
by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))
 02 Aug 2026, 12:25 (modified: 02 Aug 2026, 12:29)
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer bDeY
 Revisions (/revisions?id=rhojAeLsg8)

Comment:

Q2. Did you redefine ρ_t ? Updated Eqs. 7-9.

No: ρ_t is not redefined, and $\hat{w}$ is not a new factor folded into it. What our first response added is a definition the submission left implicit: the reference policy in the denominator for teacher samples. Once that convention is stated, the teacher term of Eq. 9 and the single corrected ratio are the *same quantity*, written with the parenthesis in two different places:

$$\hat{w}_\alpha \cdot \rho_t; =; \hat{w}_\alpha \cdot \frac{\pi_\theta}{\pi_{\theta_{\text{old}}}}; =; \frac{\pi_\theta}{\pi_{\theta_{\text{old}}}/\hat{w}_\alpha}; =; \frac{\pi_\theta}{\hat{\pi}_\beta}, \quad \hat{\pi}_\beta := \pi_{\theta_{\text{old}}}/\hat{w}_\alpha.$$

Double counting would arise only if the denominator of ρ_t in Eq. 9 were *already* the teacher behaviour policy, but it is not. The denominator is $\pi_{\theta_{\text{old}}}$ for every sample, exactly as in Eq. 7. The clarified equations are shown below:

Eq. 7: on-policy ratio (unchanged; this denominator convention holds for all samples).

$$\rho_t = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \quad \mathcal{L}_{\text{PG}}^{\text{on}} = -, \mathbb{E}_{t \sim G^o}! \left[\min! (\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \varepsilon, 1 + \varepsilon) \hat{A}_t) \right],$$

Eq. 8: the density-ratio weight, stated in full (as specified in our first response).

$$\hat{r} = \frac{D_\phi}{1 - D_\phi}, \quad \hat{w}_\alpha = \frac{\hat{r}}{(1 - \alpha), \hat{r} + \alpha} \in \left(0, \frac{1}{1 - \alpha}\right],$$

where α is the teacher fraction of the discriminator buffer, estimated online. Measured $\alpha \approx 0.10$ to 0.12 , hence $\hat{w}_\alpha \leq 1.13$: the weight is bounded and can only *down-weight* a teacher sample, never amplify one. This boundedness is what produces the fade-out reported in §4.4 and is why we use $\hat{w}_\alpha$ as a variance-controlled replay weight rather than an exact likelihood ratio.

Eq. 9: teacher term, written as a single corrected ratio. For teacher tokens the behaviour log-likelihood is unavailable, so the reference policy is imputed from the density-ratio estimate and substituted into the same clipped surrogate:

$$\log \hat{\pi}_\beta(a_t | s_t); =; \log \pi_{\theta_{\text{old}}}(a_t | s_t) - \log \hat{w}_\alpha, \quad \hat{\rho}_t = \frac{\pi_\theta(a_t | s_t)}{\hat{\pi}_\beta(a_t | s_t)} = \hat{w}_\alpha \rho_t,$$

$$\mathcal{L}_{\text{PG}}^{\text{tch}} = -, \mathbb{E}_{t \sim G^\beta}! \left[\min! (\hat{\rho}_t \hat{A}^\beta, \text{clip}(\hat{\rho}_t, 1 - \varepsilon, 1 + \varepsilon) \hat{A}^\beta) \right],$$

with the **same** ε as the on-policy term. The clip structure is unchanged from the submission: $\text{clip}(\hat{w}_\alpha \rho_t, \cdot)$ and $\text{clip}(\hat{\rho}_t, \cdot)$ coincide term by term.

We hope this resolves your questions. Thank you again for the careful reading.

Add: [Official Comment](#) [Author AC Confidential Comment](#)

Response for Reviewer bDeY, Part 1 of 2

Edit ▾



Official Comment

by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))

30 Jul 2026, 21:18 (modified: 30 Jul 2026, 21:22)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer bDeY

Revisions (/revisions?id=3F3xakB6G6)

Comment:

We sincerely thank you for the rigorous technical feedback. Your question on Eq. 9 prompted us to re-check the derivation against the implementation line by line. We also added new ablations to quantify the standalone contribution of the correction mechanisms. We address the weaknesses here and the remaining questions in Part 2.

Weakness 1: The “principled” claim, the State Channel, and Eq. 9

Do the gains come mainly from State Channel shaping?

The correction mechanisms make a substantial contribution independently of the State Channel:

1. **Baseline separation, correcting Bias 1.** Removing it causes training to collapse to **0.0%** in both environments. All teacher-mixing baselines in Table 1 also use this correction, so their reported performance depends on it as well.
2. **Corrections without the State Channel.** Removing SC entirely still gives **31.0%** on ALFWorld, compared with 5.5% for LUFFY and 1.0% for GRPO. Thus, more than 25 points of gain remain without any shaping signal.
3. **DR3, correcting Bias 2.** Removing DR3 costs 14.0 points at step 50 on ALFWorld (42.5 to 28.5) and reduces WebShop from 35.5 to **9.5**.

These results support a complementary interpretation: **the correction mechanisms are effective on their own, while SC adds further gains by extracting progress information from teacher states.**

The two components address different problems rather than substituting for one another.

Why Eq. 9 does not double-count the correction

The key is the denominator convention of ρ_t . For both on-policy and teacher samples, the reference policy is the previous student policy π_{old} , not each sample's original behavior policy. For teacher samples, we define

$$\log \hat{\pi}_{\beta} = \log \pi_{\text{old}} - \log \hat{w},$$

which gives

$$\rho_t = \frac{\pi_{\theta}}{\hat{\pi}_{\beta}} = \frac{\pi_{\theta}}{\pi_{\text{old}}} \hat{w} \approx \frac{\pi_{\theta}}{\pi_{\text{old}}} \frac{\pi_{\text{old}}}{q} = \frac{\pi_{\theta}}{q},$$

where q is the replay-cache distribution and $\hat{w} \approx \pi_{\text{old}}/q$ is estimated by the discriminator. The two factors therefore correct different terms exactly once: $\pi_{\theta}/\pi_{\text{old}}$ accounts for policy drift, while π_{old}/q accounts for replay mismatch. Their product gives the required teacher-sample weight π_{θ}/q , with no duplication. The double-counting interpretation arises only if ρ_t is read as already using the teacher behavior policy in its denominator. We will revise the paper to state the imputation and denominator convention explicitly.

Weakness 2: Evaluation breadth, adaptation beyond structured environments, and comparison with CHORD

Scope and applicability

DUET targets **multi-turn agentic tasks**, where an agent repeatedly interacts with an environment and receives a trajectory of observations. In this setting, states arise naturally from interaction, and the progress table is constructed automatically from teacher trajectories. This covers common agent settings such as web navigation, embodied household tasks, and tool use. Standard single-turn mathematical reasoning does not expose an environment-state trajectory and is therefore outside the scope of this work.

Within this scope, the Action Channel has the same input requirements as CHORD: teacher tokens, masks, and scalar rewards. It can therefore be applied wherever CHORD can, while outperforming CHORD in all four settings in Table 1 by **20.5, 24.5, 10.5, and 6.5 points**, respectively. The State Channel is an additional source of progress guidance when intermediate observations are available.

Is exact state matching required?

No. We tested this directly during rebuttal by replacing exact lookup with TF-IDF soft matching over the same progress table, without adding a learned component. We then applied 30% word dropout only to the matcher's observation input to simulate a noisy environment:

ALFWorld 1.5B, val@100 Exact matching Soft matching

Clean observations	47.5	51.5
30% observation noise	11.0	54.5

Exact matching degrades under textual corruption, while soft matching remains robust and performs at least as well on clean observations. This shows that exact matching is an implementation choice, not a requirement of the State Channel. In noisier environments, adaptation can be made at the matching interface rather than by redesigning DUET.

Add:

Official Comment

Author AC Confidential Comment



Response for Reviewer bDeY, Part 2 of 2

Edit

Official Comment

by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))

30 Jul 2026, 21:20

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer bDeY

Comment:
What is the group composition? Is m fixed, or do you resample until success?

Each prompt uses a fixed group of **7 on-policy rollouts and 1 teacher trajectory**, for a total group size of 8. Teacher trajectories are sampled without replacement from a frozen offline cache. Training itself does not interact with the environment or call the teacher model.

There is no resample-until-success procedure during training. Success filtering occurs only during offline cache construction. If a task has no cached teacher trajectory, the teacher slot is replaced by an additional on-policy rollout, so the group size remains 8. The realized average numbers of teacher trajectories per prompt are 0.978 on ALFWorld and 0.858 on WebShop.

Does SFT use the same amount of teacher data? Can you provide its training curve?

Yes. SFT uses the same cache file and sampling procedure, with a realized teacher ratio closely matched to DUET's, 0.975 versus 0.978 trajectories per prompt. The total optimization budget is also matched: **50 SFT steps followed by 50 GRPO steps**, compared with 100 DUET steps.

The SFT stage trains normally: its loss decreases from 0.84 to 0.15, while training success increases from 0.02 to 0.33. We will include this curve in the revision. We also tested a larger SFT budget by doubling the task count and increasing training to 150 total steps. Performance on WebShop decreases to 7.5%, compared with 18.5% for the configuration reported in Table 1. Thus, the reported result is already the stronger SFT configuration.

How large and diverse is the teacher cache?

For ALFWorld, the cache contains 19,497 successful demonstrations covering 2,348 tasks, or 8.3 demonstrations per task on average. The success filter retains 80.6% of raw samples.

For WebShop, the cache contains 26,178 successful demonstrations covering 5,691 tasks, or 4.6 per task. Each entry contains an environment-verified successful action sequence paired with a 72B-generated rationale.

For ALFWorld diversity, among the 8.5 demonstrations available per task on average, 4.4 have distinct action sequences, and only 7.6% of tasks contain a single solution path.

Formatting: Missing underlines in Table 1

Thank you for catching this. The revision will underline the strongest non-DUET baseline in all four columns. No numerical result or value in the Δ row changes.

Add:

Official Comment

Author AC Confidential Comment



Official Comment by Reviewer bDeY

Official Comment by Reviewer bDeY 02 Aug 2026, 08:36
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer bDeY

Comment:

Thanks for the rebuttal. I still have a few questions.

- Does SFT+GRPO in Table 1 rely on "Baseline separation"? If not, the strong performance of SFT+GRPO seems to indicate that the baseline separation is only crucial for DUET. Then, my argument is that the performance gain of DUET over existing approaches is mainly from SC, which is a good heuristic but not "principled". I am concerned that the framing and the narrative oversells the contribution.
- Did you redefine ρ_t in the rebuttal? so now w is part of ρ_t ? What is the updated version of eq 7,8,9?

Add:

Official Comment

Author AC Confidential Comment



Replying to Official Comment by Reviewer bDeY

Official Comment by Authors

Edit ▾

Official Comment
by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))

02 Aug 2026, 12:24
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer bDeY

Comment:

Thank you for the follow-up. Both questions deserve exact answers, and the first one helped us see what our framing should have said all along.

Q1a. Does SFT+GRPO rely on baseline separation?

No, the RL stage is pure on-policy GRPO, so no teacher trajectory enters any rollout group. Your premise is therefore correct, and we would like to state its consequence precisely, because we believe it supports the paper's diagnosis rather than undermining it:

Baseline separation is not "crucial only for DUET", it is crucial for any method that mixes teacher rollouts into the GRPO group. Bias is created by the act of mixing. SFT+GRPO avoids it by consuming the teacher once, offline, and consequently has no teacher signal at all during RL. Its immunity is not a counterexample to our framing; it is the same claim seen from the other side that mixing is not free.

Method	Teacher in GRPO group?	Baseline sep. active?	AF-1.5B	WS-1.5B
On-policy GRPO	no	n/a (nothing to separate)	1.0	0.5
SFT → GRPO	no (either stage)	no	30.0	18.5
LUFFY	yes	yes	5.5	5.5

Method	Teacher in GRPO group?	Baseline sep. active?	AF-1.5B	WS-1.5B
DUET w/o baseline sep. (keeps SC + BC + DR3)	yes	no	0.0	0.0
DUET	yes	yes	47.5	36.0

For mixing methods the correction is necessary but not sufficient: DUET without it collapses to 0.0, while LUFFY reaches only 5.5 even with it enabled.

Q1b. Is DUET's gain then mainly from SC?

The State Channel is useful, it densifies a sparse reward, and on ALFWorld-1.5B it contributes the largest share of the final margin; the revision will state this plainly. But **it is not the sole source of DUET's gain, because DUET is built to use everything a teacher rollout contains**: the actions and reasoning it demonstrates (behavior cloning and corrected replay) as well as the states it visits (SC). With SC removed entirely, the corrected core still reaches 31.0 on ALFWorld, on par with SFT+GRPO (30.0) and more than 25 points above LUFFY (5.5). And on WebShop the attribution inverts: removing DR3 alone, with SC held fixed, collapses DUET from 36.0 to 9.5.

The corrections also carry independent value of exactly the kind we call principled. Baseline separation removes the systematically negative advantages that successful teacher rollouts impose on on-policy samples through the shared group baseline. Without it, training collapses to 0.0 in both environments *even with SC, BC, and DR3 all retained* (shown in the table above), so SC's contribution is not even realizable on an uncorrected mixer. Density-ratio repair attenuates the teacher's influence adaptively, tracking the estimated teacher-student gap rather than a hand-tuned schedule, and is bounded ($\hat{w}_\alpha \leq 1.13$) so it can only down-weight. These two corrections are the principled part of the contribution.

That said, we agree with you that folding SC under the same "principled" banner overclaims. We will rescope the narrative to present DUET primarily as a **bias-corrected experience-replay algorithm**, with SC as a clearly labeled heuristic extraction channel.

Add: [Official Comment](#) [Author AC Confidential Comment](#)

Official Review of Submission32282 by Reviewer UyKJ

Official Review by Reviewer UyKJ 📅 24 Jun 2026, 15:16 (modified: 03 Aug 2026, 10:48)
👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer UyKJ
📄 Revisions (/revisions?id=aBExPGxLKf)

Summary:

This paper first points out that training small LLMs with RL is difficult because a weak student usually fails on its own, so it suffers from the cold-start problem that it gets almost no reward to learn from. Then this paper introduces two failure modes under the common fix of naive teacher experience replay in GRPO. Motivated by this, this paper proposes Dual-Channel Teacher Utilization (DUET), which splits each teacher trajectory into two separate channels, namely the Action Channel and the State Channel. The Action Channel is able to provide corrected action-level supervision, and the State Channel can give the student dense rewards for making teacher-like progress on its own. Empirically, DUET beats all baselines in every setting, even when the biggest gains on the weakest students cold-start is the worst.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

- The idea of this paper is clear and novel. The failure modes are clearly diagnosed. Each proposed component maps cleanly onto a specific failure mode, which makes the design feel clean.

- The two teacher fade-out paths are smart, in that the teacher influence recedes through reward progress and distributional progress.
- The ablation study is thorough, making the main claim convincing.

Weaknesses

- Although the results are strong, the experiments only cover 2 envs, 1 family of LLM, and a single teacher, so I think the generalizability might be questionable. The pipeline also relies a lot on the quality of the teacher's experience. And the training seems hard because there are many hyperparameters to tune.
- One notable exception is the pick-two task at 3B, where GRPO outperforms all teacher-mixing methods. This suggests that teacher demonstrations may not always be beneficial for tasks requiring broad combinatorial exploration. I feel the reason behind the exception could be explained more clearly.
- DR3's theoretical justification is not strong enough, and the authors themselves call it a bias mitigating replay weight rather than an exact likelihood ratio, because the teacher cache is success-filtered.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:

1. The evaluation is limited, so could the authors provide any evidence that DUET can generalize to a different teacher or a non-Qwen student? And how does performance change when the teacher is weaker?
2. Could the authors explain in more detail what might cause the exception in the Pick Two task?
3. The teacher's data are all from successful trials, so how do the authors know that the discriminator is separating teacher from student, and not just success from failure?

Limitations:
yes

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
NA. This should be fine.

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Add: **Official Comment** **Author AC Confidential Comment**



Official Comment by Authors

Edit

Official Comment
by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))
 03 Aug 2026, 20:17
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer UyKJ

Comment:
Dear Reviewer UyKJ,

As the discussion period is coming to a close, we would be grateful if you could review our response and the additional evidence provided. We have addressed each concern raised in your review; if any questions remain, we would sincerely appreciate your feedback and the opportunity to clarify them before the discussion ends. Thank you again for your time and thoughtful review.

Add: **Official Comment** **Author AC Confidential Comment**



Official Comment by Reviewer UyKJ

Official Comment by Reviewer UyKJ 03 Aug 2026, 20:42
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer UyKJ

Comment:
Thank you for the detailed response and additional experiments. They addressed my main concerns. I will maintain my current score and have increased my confidence in my assessment of the paper.

Add: Official Comment Author AC Confidential Comment



Response to Reviewer UyKJ, Part 1 of 2

Edit

Official Comment
by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))
 30 Jul 2026, 20:47
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer UyKJ

Comment:
We sincerely thank you for the positive assessment and the thoughtful questions. During rebuttal, we ran three targeted experiments: a teacher-quality ablation from 72B to 14B, a cross-family experiment with a Llama-3.2-3B student and Qwen teacher, and a direct probe of what the discriminator separates. We address the weaknesses here and the remaining questions in Part 2.

Weakness 1: Dependence on teacher quality and tuning complexity

Teacher quality

DUET studies how to use stronger-teacher experience effectively, so access to a teacher is part of the problem setting. The key question is whether the utilization mechanism remains effective as teacher quality decreases. We therefore replaced the 72B teacher with a substantially weaker 14B teacher while keeping all other settings fixed:

ALFWorld 1.5B, val@100 72B teacher (80.6% success) 14B teacher (68.1% success)

DUET	47.5%	34.5%
CHORD with 72B teacher	27.0%	—
LUFFY with 72B teacher	5.5%	—
GRPO without teacher	1.0%	1.0%

Despite a 12.5-point drop in teacher success, DUET with the 14B teacher still outperforms every baseline using the stronger 72B teacher. The 14B run is also still improving at the end of the fixed budget, indicating graceful degradation rather than failure.

We observe the same pattern when the student becomes much stronger. With a 7B student, DUET reaches 86.5%, compared with 85.0% for GRPO, while LUFFY falls to 82.5%. Thus, when teacher experience offers little remaining advantage, DUET reduces its influence rather than allowing it to become harmful.

Hyperparameters

DUET replaces several manually tuned schedules with adaptive quantities:

- the BC coefficient μ is determined by discriminator accuracy rather than a fixed decay schedule;
- teacher influence decreases automatically as the student improves;
- the channel coefficients and DR3 settings are shared across all main-table configurations.

The only environment-specific quantity is the BC cold-start threshold d_floor , which is set once before training.

Weakness 2: The Pick-Two exception

We agree that this exception deserves a clearer explanation. It arises from a specific limitation of the current potential construction. Our Φ assigns each state the maximum normalized position at which it appears in a demonstration. Pick-Two revisits states much more often than other task types: 52.6% of its demonstrations contain repeated states, compared with 7.5–30.8% elsewhere. The sequence “put down item A, then find item B” naturally returns to earlier states, causing Φ to saturate during the first sub-goal and provide little discrimination for the second.

This explains why the exception is localized to Pick-Two rather than reflecting a general failure of teacher guidance for combinatorial exploration. DUET still outperforms GRPO on Pick-Two at the other two scales: 24.4% versus 0.0% at 1.5B, and 77.8% versus 75.6% at 7B. A direct remedy is to condition Φ on the current sub-goal. We will add this analysis and limitation to the revision.

Weakness 3: Theoretical justification of DR3

Discriminator-based density-ratio estimation has a standard theoretical basis: under a Bayes-optimal discriminator, $D/(1 - D)$ recovers the density ratio. Two design choices adapt this idea to our success-filtered replay setting.

1. **The ratio is defined against the replay distribution.** The relevant sampling source is the success-filtered cache, not the unfiltered teacher policy. The discriminator is therefore trained to distinguish cache samples from current-policy samples, matching the distribution that actually enters the update.
2. **The applied weight is smoothed and bounded.** We use $\hat{w}_\alpha = \hat{r}/((1 - \alpha)\hat{r} + \alpha)$, which approaches $D/(1 - D)$ as $\alpha \rightarrow 0$ while bounding the weight by 1.13. Teacher samples can therefore be down-weighted but never amplified. This is why we describe DR3 as a bias-mitigating replay weight rather than an exact, unconstrained importance ratio. We will state this form explicitly in the revision.

DR3 also has a clear empirical contribution. Removing it reduces ALFWorld performance at step 50 from 42.5% to 28.5%, a 14.0-point drop. Its main benefit is faster and more stable use of replayed teacher experience, not only a higher final score.

Add:

Official Comment

Author AC Confidential Comment



Response to Reviewer UyKJ, Part 2 of 2

Edit ▾



Official Comment

by Authors (Shafiq Joty (/profile?id=~Shafiq_Joty1), Wenya Wang (/profile?id=~Wenya_Wang1), Qisheng Hu (/profile?id=~Qisheng_Hu1), Quanyu Long (/profile?id=~Quanyu_Long1))

30 Jul 2026, 20:55

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer UyKJ

Comment:

Can DUET generalize to a weaker teacher or a non-Qwen student?

For a weaker teacher, the new ablation in Weakness 1 provides direct evidence. Reducing teacher sample success from 80.6% to 68.1% lowers DUET from 47.5% to 34.5%, but the weaker-teacher result still exceeds every baseline using the strong 72B teacher. With a 7B student, where teacher experience offers little remaining advantage, DUET still reaches 86.5% versus 85.0% for GRPO, while LUFFY drops to 82.5%. Together, these results show graceful degradation as teacher quality decreases.

For cross-family transfer, we ran a new experiment using a **Llama-3.2-3B student with the unchanged Qwen-72B teacher cache**. The only adaptation was setting the SC coefficient once to match the new student's initial reward scale. Training remained stable, and DUET

reached **15.0%**, compared with 5.5% for no-teacher GRPO. Thus, Qwen teacher trajectories can effectively train a Llama student, providing evidence that DUET transfers across model families.

What causes the Pick-Two exception?

As discussed in Weakness 2, Pick-Two demonstrations revisit states much more frequently than other task types. Because the current potential assigns each state its maximum normalized position, repeated states cause the potential to saturate during the first sub-goal and weaken guidance for the second. This is a localized limitation of the potential construction rather than a general failure of teacher guidance.

Does the discriminator separate teacher from student, or success from failure?

Because the cache contains only successful trajectories, we tested this concern directly. If the discriminator mainly captured success, restricting the student side to successful trajectories should make the two classes harder to separate, since both sides would then contain only successes.

The opposite occurs:

- 1. **Separability increases when only successful student trajectories are used.** Probe accuracy rises from 84.7% to 90.0% on ALFWorld, and from 94.0% to 99.0% on late-stage WebShop.
- 2. **Successful and failed student trajectories receive nearly identical scores.** Their average B(sreview.net/getting-0.861 and 0.869, compared to 0.800 for the group started/frequently asked-questions) id=OpenReview.net/Support)
- 3. **The late-stage WebShop result provides the strongest test.** The student already succeeds on 84% of rollouts. A success detector should therefore assign successful student trajectories more teacher-like scores. Instead, they appear more student-like than failed trajectories (0.947 versus 0.904).

These results indicate that **the discriminator captures policy identity rather than task outcome**.

OpenReview (/about) is a long-term project to advance science through improved peer review with legal nonprofit status. We gratefully acknowledge the support of the OpenReview Sponsors (/sponsors). © 2026 OpenReview

Add: Official Comment Author AC Confidential Comment